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Well Duration Taskforce: A Precision Strike on the Dragging Factors Impacting Well Delivery

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Abstract

The paper aims at showcasing the multifaceted approach that facilitated Well delivery recovery from 90% in July'24 to 102% by Dec'24 year for a drilling team handling one of the largest brown fields in Middle East and North Africa (MENA) Region. The Well delivery taskforce created an impact by reduction in well duration and NPT through optimization of operations, embedding new technology, reintroducing past success and preserving consistency. Key focus of the paper includes establishment of challenging performance benchmarks, review of well designs, revamping of drilling programs and practices, recycling of lesson learnt into drilling/completion programs and to increase technology footprint that targets well-duration reduction. From present study, 37% of action items deployed delivered duration savings from the first well thereby ensuring a psychological push to the whole new approach. 30% of the initiatives showed the impact of ensuring consistency in operations and reduction in NPT. Additionally, well recovery rose from 90% to 96% with a reduction in average well duration by an impressive 6%. The overall Invisible Loss Time (ILT) on drilling rigs reduced from 34.1% to 33.9% and 31.4% in Q3 and Q4 respectively compared to Q2. The well duration taskforce brought together a new thought process to challenge Best of Best in each sub activity, wherein overruling the conventional P75 approach. This brought about momentum to the operations that is leveraged on resilient consistency.

Keywords: Well duration reduction, Innovative practices, NPT reduction, Process Improvement, Technology induction

Introduction

An inevitable part of engineering challenge over the years have been the increased drilling complications and well trajectory complexities associated with a brown field due to depleted reservoirs and anti-collision issues. Vintage issues including severe mud loss situation in 12.25" section in aquifer reservoirs have predominantly shown an increased severity that was affecting the average Rate of Penetration (ROP). Challenges associated with water wells and water haulage have not been able to support well operations

as much as it supported in the past. Over the years, focus on high drain hole length have gained popularity owing to the favourable production economics associated with Maximum Reservoir Contact (MRC) wells. Failures associated with directional drilling tools while drilling such wells have shown an increasing trend. The failures were mostly attributed to high shocks and vibrations experienced while drilling these sections. The sub-surface reservoir fault characteristics of the field have high throws and mud loss characteristics, otherwise not prognosed during well planning. These have led to severe mud loss conditions and well activity. Requirement of sophisticated Logging While Drilling (LWD) tools have significantly impacted the practical ROP in the wells.

Back-reaming severity against the aquifers have impacted performance in majority of the wells in the field and has been a legacy issue that showed minimal results with the countermeasures attempted over the years. The demand for horizontal hole equipped with sophisticated lower completion equipment have increased the complexity of the well operations. There have been several cases of non-productive days associated with lower completion issues.

The inclusion of new technology and practices have been able to offset the impact of the challenges to some extent (Li et al., 2022; He et al., 2023). There has been a significant change in drilling bit philosophy and availability of custom Polycrystalline Diamond Cutter (PDC) bits that have shown promising results in improving the on-bottom ROP. The inclusion of Anti Stick slip tools in directional BHA (Bottom Hole Assembly) has also claimed savings but lacked a definitive data-based confirmation. Enablers like the Triple automated catwalk and bucking unit have provided much needed support in reducing the flat time associated with tubular make up and lay down. Apart from other best practices, these methods have shown promising time saving impact, however there had neither been significant research on a micro level nor scaled up for implementation at function level in many cases. In 2024, a case study on average duration of a light casing design well (2 casing + 1 Liner policy) with an average horizontal hole length of 3000ft with lower completion was found to be 32.56 days (Fig. 1). In 2019, a well was completed with fastest well duration of 20.6 days (25.8 days when extrapolated to comparable well requirements). This comparison outlined a significant scope for improvement by analysing the detrimental factors affecting well duration, address them with best practices / technology, quantify the savings and upscale for further savings across the field (Levorato et al., 2023).

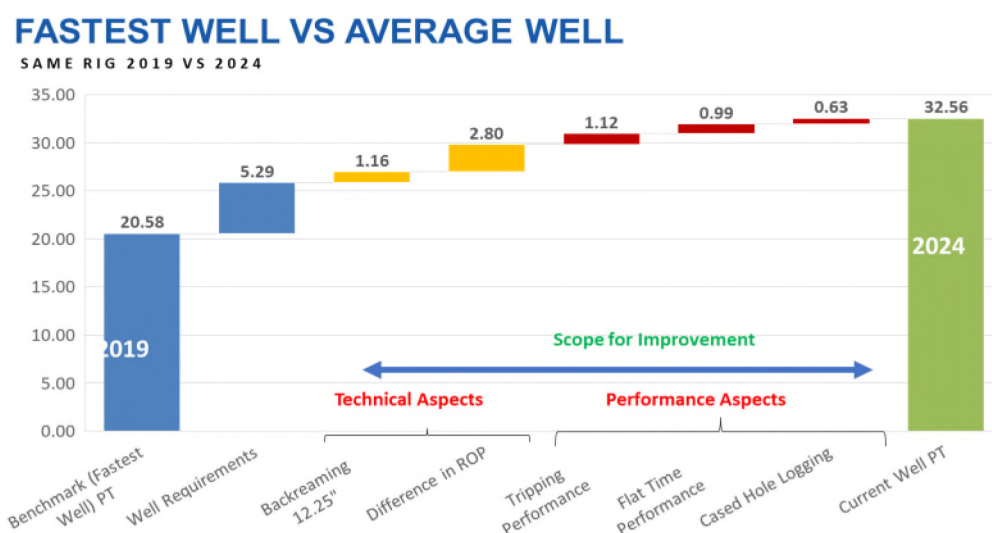


Figure 1—Comparison of Fastest well in 2019 vs Average well in 2023-24

The present study aims at showcasing the multifaceted approach facilitating well delivery recovery from 90% in July'24 to 102% by Dec'24 year for a drilling team handling one of the largest brown fields in MENA

Region. The target for enhancing well delivery was required to be achieved maintaining the same well mix and no additional added rig years. The challenge was hence focussed on methods to achieve a better performance from each rig for the remaining half of the year. This required a strong focus on average well duration and a strategic plan of action to reduce it. Hence, the target was to reduce average well duration by optimization of operations, embedding new technology, reintroducing past success and maintaining consistency amongst the fleet of different rigs. Key focus areas were evident from the preliminary analysis and required attention to both engineering and ILT factors that had scope for improvement.

Methodology

Traditional methods of performance analysis have relied upon performance benchmarking with similar well types. Three prominent well designs deployed in the field includes Heavy Casing Design, Light Casing Design & Liner less Design. Based on target reservoir and well functionality, there are 14 prominent well types utilised in the field. Categorisation based on these wide variety of well types significantly reduced the database quality. A varied approach associated with drilling sub-activity level analysis was chosen to ensure all drilling operations wherever applicable were treated as comparable irrespective of well type (Tian and Dheyaa, 2025). This provided a broader database outlook enabling focus on pinpointing the dragging factors embedded in these operations. Sub-activity level analysis is crucial to identify the top performers enabling a review of productive factors that lead to maximised performance.

Real time data of more than 143 wells were analysed to identify clean data and secure sub-activity wise breakups for each well. The strategic approach is to understand that rigs in the fleet are of varied capacity and technical capabilities that would not provide a biased result if average values are considered for analysis. Averages tend to diminish the impact of high-performance data and mask the low performance indicators. A realistic perspective that could be beneficial was the percentile approach that provide insight into the percentage of data in a dataset that fall below a target data point. This gave a performance driven target aimed at reviving dragging sub-activities. To categorise logically and derive fruitful analysis the following percentiles were considered for each sub activity analysis; P10, P25 and P50.

Sub-activity Analysis

The drilling sub-activity for a regular LCD well can be sub divided into 61 items each corresponding to a distinctive measurable duration that is methodically planned at the beginning of each well by means of Drill Well On Paper (DWOP). During the actual drilling operation, each sub-activity is precisely quantified based on actual real time data and recorded. Sub activities are classified on a drilling phase wise basis to effectively monitor impact at each milestone (Fig. 2)

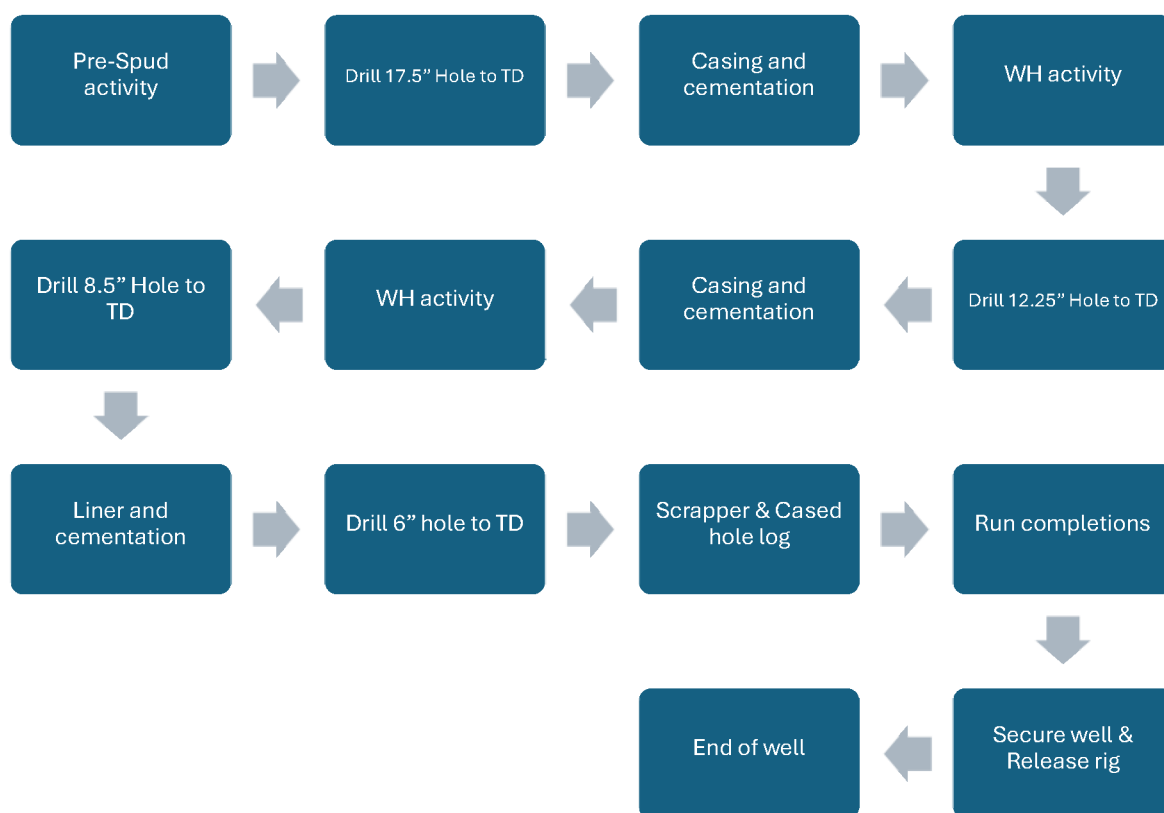


Figure 2—Drilling Sub-activity (not sub-categorised) for a well

For the period 2019 to 2024 (Fig. 3), an analysis was carried out to identify the percentile variance of high performing sub activities to low performing sub activities. This provided an insight into the key dragging sections thereby allowing a focussed approach to identify the factors resulting in the low performance drilling sub activity across the fleet. The data re-confirmed the results of initial data analysis carried out on fastest well and average well.

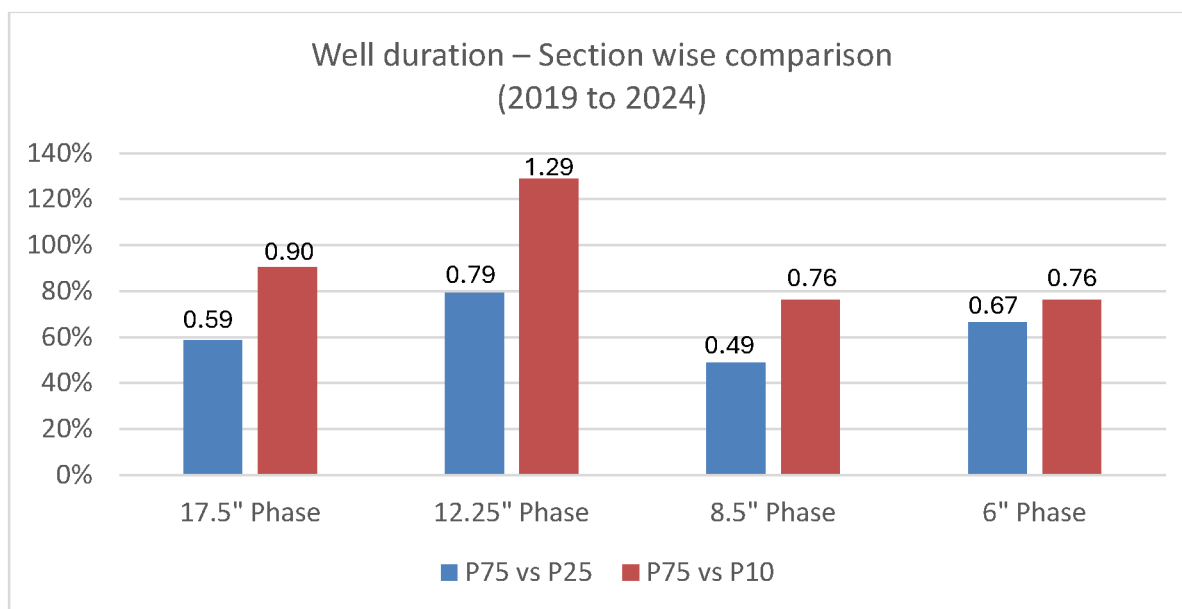


Figure 3—Well duration -Section wise comparison

The relative comparison between P75, P25 and P10 provides a comparative analysis to understand the phase that was highly suitable to be impacted by interventions. A higher difference between P75 vs P25 and P75 vs P10 gives an indication that there is a high variance between the top performing durations and the majority. This provides a clear indication of low performance areas. As per the data it is clear that the maximum impact was possible if the issues in 12.5" section are addressed. Accordingly, prioritisation of targets was worked upon and charted.

A tailored well based on P10 values over the period 2023-24 re-affirmed that sub 20 days duration well (without lower completion) was achievable (Fig. 4).

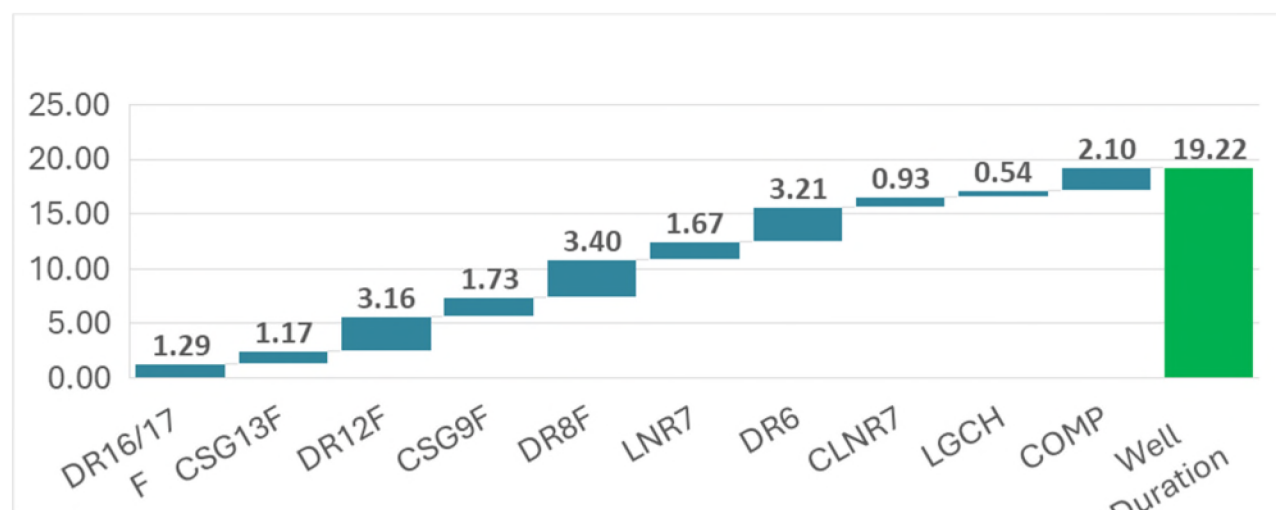


Figure 4—Tailored well based on P10 during the period 2023-24

Action plan

Based on the targeted phase and operations, the game plan was strategized to implement quick fixes and shortterm action items. The action items were carefully chosen to ensure that there was quantifiable impact on the well duration. The action items were implemented on focus well and slowly scaled up to other wells/rigs (Fig. 5).

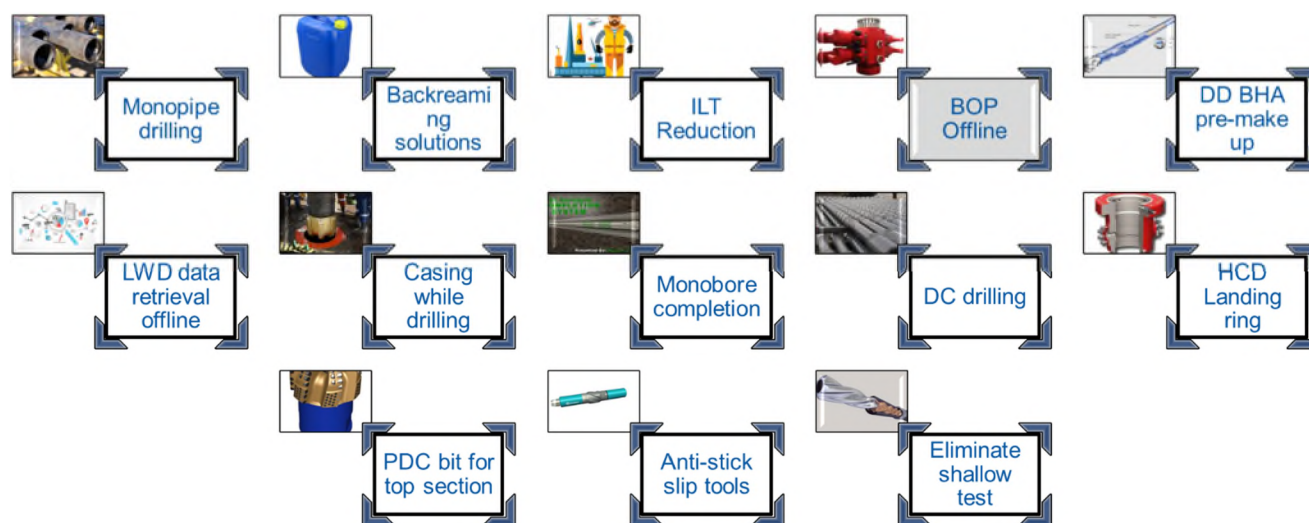


Figure 5—Initiatives deployed

Mono pipe drilling

The concept revolves around utilisation of 4" drill pipe in 8.5" deviated section in place of conventional 5.5" drill pipe. The advantage is based on the optimisation on account of time saving for the pickup and lay down of the conventional 5.5" drill pipes that would be utilised only for drilling the 8.5" deviated section. Extensive simulation and analysis were carried out inhouse to preselect wells that were conducive for monopipe drilling without hydraulic or torque associated risks. The average cost saving works out to approximately 12 hours each well.

Back reaming solutions

The impact of back reaming against aquifers in the 12.25" section accounted for 6-24 hours in the wells. The utilisation of high-performance mud system equipped with fibres, drilling with downhole mud motor with bend angle, utilisation of bi-directional or unidirectional reamers in the BHA have been implemented on case to case basis based on offset well data. An innovative drill and seal technique has also been deployed that aims at utilising only stabilisers in BHA in place of reamers. It enabled plastering effect of the drilled cuttings against the highly permeable/porous aquifers. Drilling was also carried out using plain water in selected cases to avoid thick mud cake formation against the reservoir. Reduction of back reaming has not been a 100% success but has shown reduction in back reaming by more than 60%.

ILT reduction

A lower flat time efficiency accounted for more than 75% of the Invisible Loss Time (ILT) during the well progress. The micro KPI analysis provided an insight into the micro KPIs that required focus. This included section wise analysis on the slip to slip connection timing, cased hole tripping speeds, directional survey time, Wellhead work time, BHA make up time to name a few. A detailed ILT booklet embedded with specific operation wise targets for each operation enabled the rig team to have micro targets.

Directional Drilling

A varied and diverse range of flat time associated with Directional drilling BHA and programming was observed across the rigs. Standardisation of few critical steps involved in directional BHA and operations were focussed and upscaled to cover the fleet. This included maximisation of BHA component makeup in the service company yard with only limitation to suit the trailer box length. Additionally, DD tool programming and memory data downlinking was carried out completely offline.

Based on data of shallow tests carried out on several service provider it was with more than 99.9% confidence that shallow test never gave negative results and hence were avoided altogether resulting in significant time saving on each well.

Surface hole drilling optimisation

Drilling of the surface section faced high flat time due to BHA accommodation trip and lower average ROP. Based on a simulations and revised BHA strategy, a slim BHA was trialled and deployed that required lesser BHA components and hence lower makeup time. A custom-made cross over was also utilised to enable the drilling of top section without stoppage for pull out to space out the BHA. Multiple drilling dynamics analysis was carried out to utilise a carefully selected PDC drill bit of 6 blade 19mm configuration to achieve consistent best of best ROP in the top section.

New technology induction

Utilisation of Anit-stick slip tools of various designs were tried out in wells to ascertain the impact on average ROP. It was observed that longer horizontal section wells showed a 10-12% increase in average ROP owing to ability to provide full parameters as a result of reduced stick slip. All major anti stick slip tool designs provided by leading technology suppliers were utilised. The position of AST tool was also bought

nearer to the RSS without impacting the LWD functionality and showed promising results. The AST tools resulted in average savings of 16 hours in a MRC well.

Casing while drilling was deployed to counter severe loss regions to carry out 9-5/8" casing job. Due to the low annular velocity requirement using casing while drilling, fluid loss was significantly manageable at 250GPM as compared to conventional drilling that had a high discharge requirement of approximately 500 GPM as bare minimum. The cement bond log against thief zones also showed an improved results as compared to a conventional drilling and casing job. The results can be attributable to the low annular disturbance while drilling leading to a better gauged hole with a casing while drilling job. The Casing while drilling job resulted in an average saving of 1.5 days per well.

For completion requirements, in place of a conventional lower and upper completion job, a monobore approach was adopted for candidate wells. The extensive simulation analysis provided the necessary confidence to attempt such completion in horizontal holes upto 5000ft. It also ensured that the completion was able to reach TD without buckling issues due to the high friction factors experienced in some wells. The use of mono bore completion restricted utilisation of assisting tools like swivels or heavy lower completion running BHA. The average saving on account of monobore completion was 28 hours in a 3000ft HH well.

Results and Discussions

The induction of best practices, new technology and new approach to performance benchmarking induced a performance breakthrough in the fleet ([Fig. 6](#)). The key contributing factor was the ability to quantify and track the initiatives phase by phase, well by well with the help of real time operations monitoring system. There has been improvement observed each quarter on performance indicator KPIs compared to previous quarter. The improvement in average ROP by 25% had an impact on not only the average well duration but also on the well cost. The induction of new technology including the Anti stick slip, new bit designs, drill collar drilling, DP offline make up to name a few, while contributing to duration saving proved to be saving on cost as well. This facilitated upscaling of the technology to all the rigs in the fleet. Flat time KPIs like well head works that had shown a regressive behaviour over the years when focussed upon as micro KPI generated considerable savings to tune of 16% and was impactful on all wells irrespective of rig type. These were facilitated using the increased focus on offline jobs for well head works. This involved deployed of additional manpower to carry out BOP offline pressure testing, offline nipple down jobs during cement jobs to name a few.

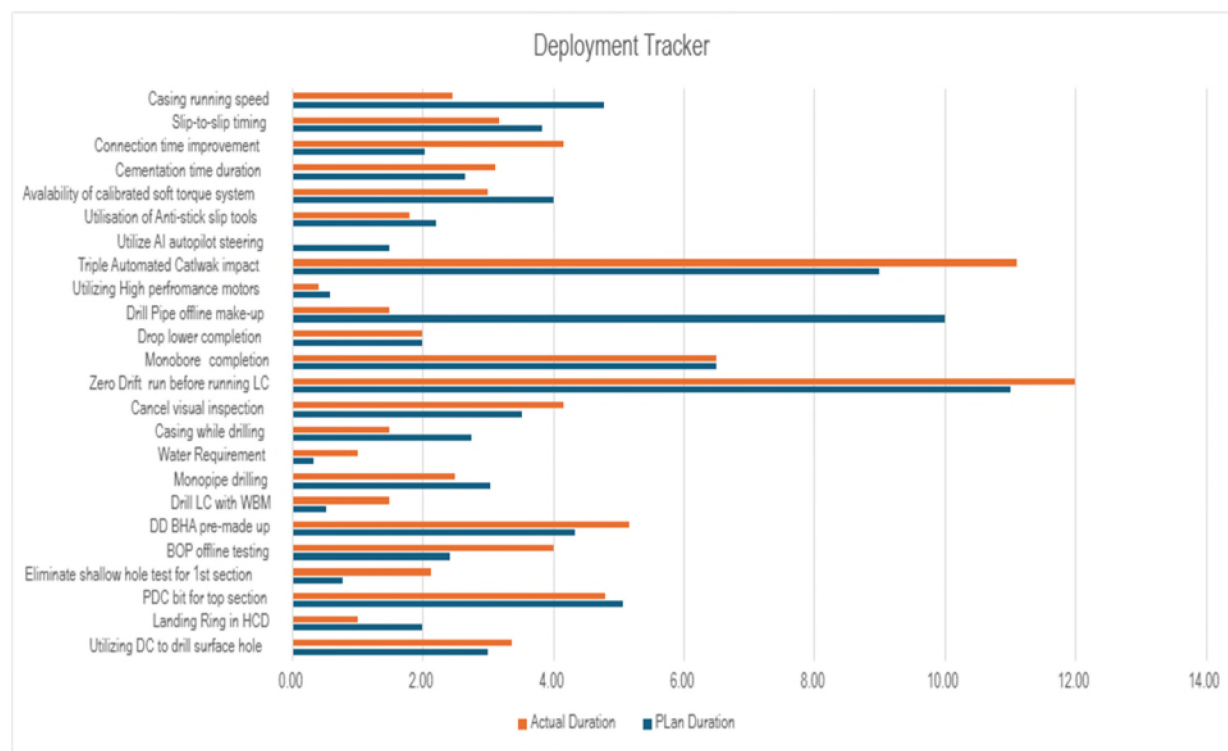


Figure 6—Action tracker for Initiatives deployed

Connection time is a major contributor to the well duration. Irrespective of being the drilling connection time of tripping connection time, as a cumulative it accounts for 9% of the total well duration. By ensuring the corrective actions on the operational processes that were delaying each connection. An improvement to tune of 26% in drilling connection time (weight to weight) of which 4% was saved in slip to slip connection time itself. Reaming / back reaming norms were reviewed based on field experience that had a significant impact on the pre-connection timings. Back reaming was considered as a legacy issue and reaming back reaming while drilling each stand was being religiously followed by all rigs. A detailed review on few well was carried out to wherein back reaming and reaming was minimised while drilling and the impact on the final pull out was quantified. It was worthwhile to observe that the impact on back reaming remained same in both cases. This provided a much-required practical back up to the concept that back reaming condition was less of a well tortuosity, rather a time linked phenomenon.

Directional Survey process was reviewed in coordination with the directional drilling service provider and an approach was taken to ensure minimum waiting was prior to resuming drilling. A generalised protocol was communicated to the rig team to commence drilling after the minimum required stationary survey time and not to wait for the complete survey to surface. This practice gave more than 99% successful survey with a saving of approximately 1 min each stand drilled.

Casing jobs and operational sequence were revamped to include optimised process for connection makeup speed and utilisation of higher speed draw works gears. Triple automated catwalks were deployed to run casings premade up using bucking unit in doubles.

The overall impact on average well duration was as expected based on the improvement in KPIs (Fig.7). The average well duration in July 2024 was 34.9 days compared to business plan for the same period planned at 31.4 having a variance of 3.5 days as shown in Fig. 8. This was crucial to the well delivery mandate and overall performance outcome. The inclusion of the initiatives resulted in a gradual improvement over the months and concluding the year at a variance that was reduced from 3.5 days in July 2024 to 1.6 days. It is noteworthy to mention that this reduction in variance equates to approximately 8 wells added.

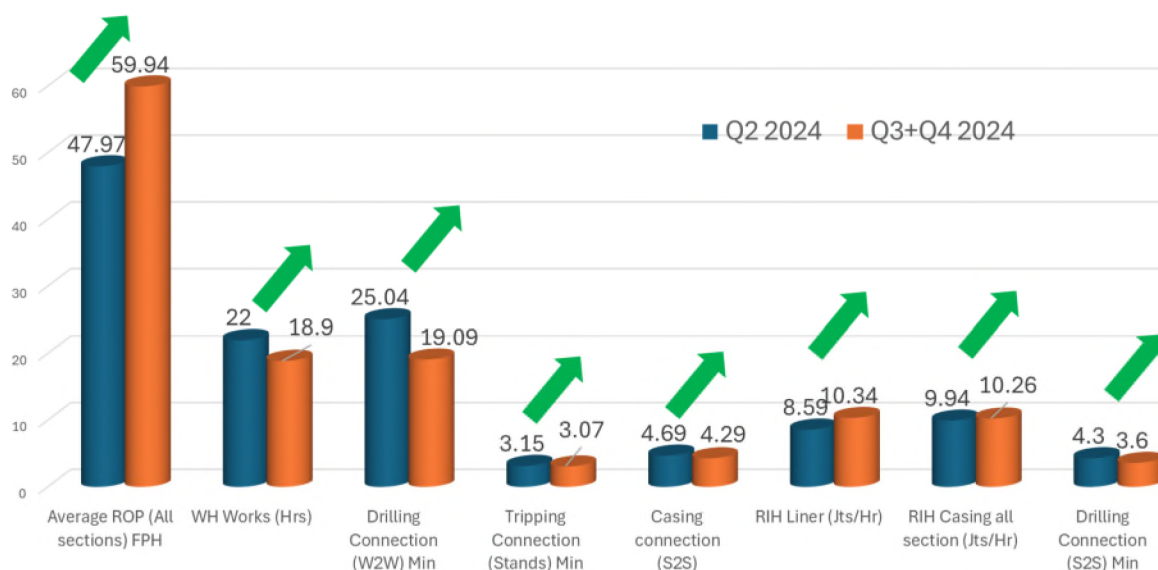


Figure 7—Impact of well duration initiatives in Q3&Q4 compared to Q2

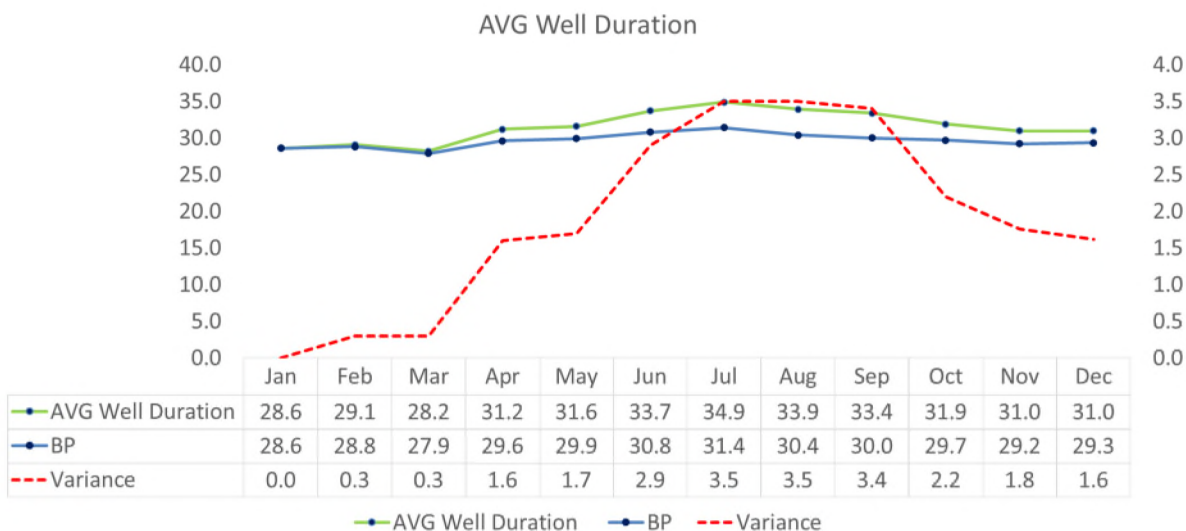


Figure 8—Progress of Well delivery and average well duration in 2024

The overall performance and well delivery improved from negative 4% to positive 2% by end of the year 2024, over achieving the recovery plan by 4 wells (Fig. 9).

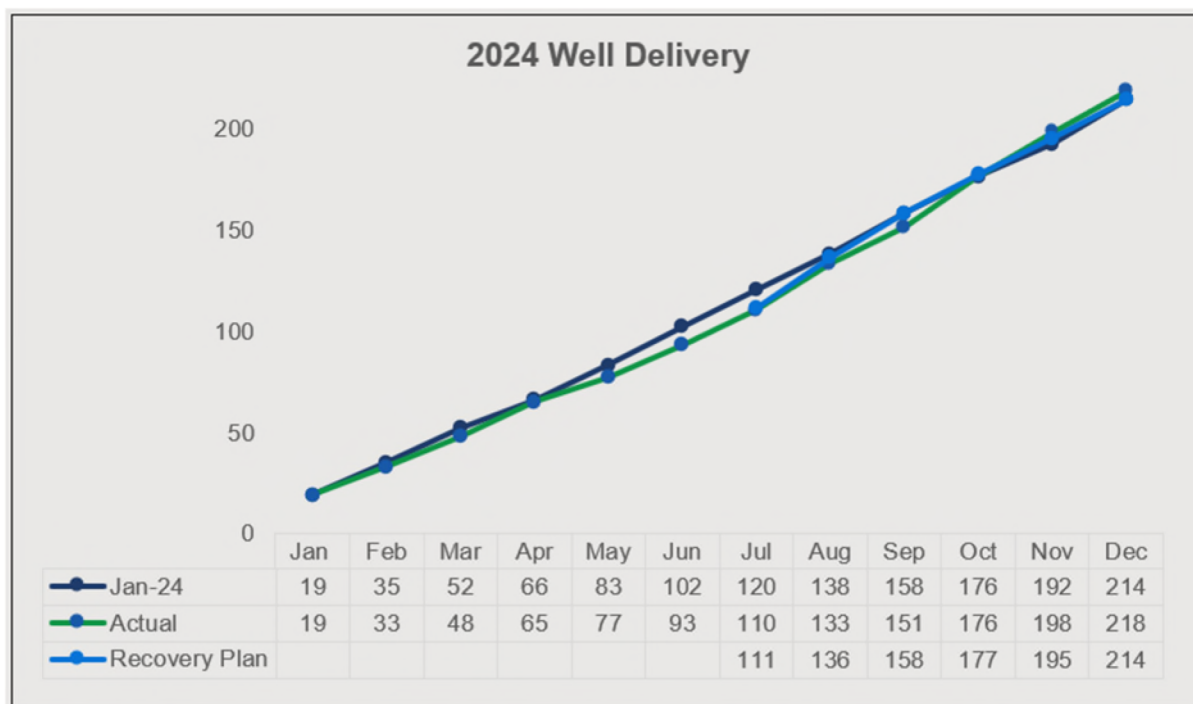


Figure 9—Final well delivery for year 2024

Conclusion

The drilling cycle in any company is highly impacted by Invisible loss time that overshadows the positive impact of technology infusion. There has been an increased necessity to carry out detailed risk assessment and management of change protocols while carrying out changes to routine operational sequence and processes. Such changes to normal process have an inherent inertia against change amongst the other stakeholders. Their concerns need to be addressed and a conducive solution can be arrived at for mutual benefit.

The initiatives deployed as part of well duration taskforce has broken many Best of Best KPIs for the field and region. As of July 2025, with the momentum imparted by the initiatives a total of 52 Best of Best records has been made. The key aspect involved rigorous tracking and follow ups at ground level to top management alike with weekly and biweekly follow up meeting.

A total of 32 focus items developed based on Performance Reliability, Equipment Availability, technology infusion and Innovative Practices were envisaged and 24 of them were practically deployed and proven successful. The balance 8 are under technical review and requires more detailed risk analysis prior to pilot deployment. Each action item was percolated to rig level for deployment and progress tracking at each phase of well drilling. The most impacted well activities by virtue of numbers were on account of renewed completion philosophy and utilisation of offline tubular handling system in addition to surface hole drilling practices and flat time operations saving contributing to savings small and large in 49 wells amongst 98 wells in Q3-Q4 combined.

The 37% of action items deployed delivered duration savings from the first well thereby ensuring a psychological push to the whole new approach. 30% of the initiatives showed the impact of ensuring consistency in operations and reduction in NPT. After 3 months of deployment the well recovery rose to 96% from 90% with a reduction in average well duration by an impressive 6%. The overall ILT on drilling rigs reduced from 34.1% to 33.9% and 31.4% in Q3 and Q4 respectively compared to Q2.

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